

Complete Test Report

FMI Skid Demo — model quality, scenario behaviour, evidence, and an honest reading

What was tested, how it performed, and — stated plainly and up front — exactly what the results do and do not prove. Written so a non-specialist can read every number with confidence.

Document	Verification & Test Report
Product	Jneopallium Industrial Loop Guardian (FMI Skid Demo)
Model	industrial-loop-guardian 1.0.0-reference-maintenance-energy
Test scope	Model quality, scenario behaviour, safety boundary, scale
Result	All checks passed (reference skid corpus)
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1 Executive summary of results

The Industrial Loop Guardian was verified at four independent levels: **model quality** (the four diagnostic heads separate each fault from normal operation), **scenario behaviour** (the running demo makes the right calls across nine plant scenarios), **acceptance checks** (the OPC UA / MQTT / PLC safety boundary holds), and **scale** (the training pipeline holds up at a 100 GiB logical-corpus target). Every check passed.

Test layer	What it checks	Result
Model quality	Per-finding precision / recall / F1 / FP rate	macro-F1 $\approx$ 0.9995 , near-zero FP
Scenario behaviour	Nine deterministic plant scenarios	Pass — diagnosis + safety as expected
Acceptance	OPC UA / MQTT / PLC boundary	Pass — boundary holds in every scenario
Scale	100 GiB logical-corpus target & guardrails	Pass — target reached, fitted sample bounded

The single most important sentence in this report

The near-perfect scores below are PIPELINE evidence on a deterministic, synthetic skid corpus — they prove the system is wired correctly and the four faults are cleanly separable. They are NOT a claim of real-world accuracy, which must be earned on external historian, CMMS, and energy-meter data. We state this here, at the top, on purpose.

2 What we tested, and why

A supervisory product earns trust by being tested the way it will be used — not just "does it run", but "does it diagnose correctly, stay inside the safety boundary, and explain itself." The verification is therefore layered:

- **The trained heads** — given windows of plant behaviour, does each head separate its fault (pump wear, oscillation, sensor drift, energy deterioration) from normal operation, and are its false positives within budget?
- **The running system** — across nine realistic scenarios, does it diagnose correctly, while the deterministic controls keep the plant safe?
- **The safety boundary** — do OPC UA, MQTT, and PLC/SIS keep their roles in every scenario, including outages and interlock trips?
- **The pipeline at scale** — does training stay correct, bounded, and reproducible at a 100 GiB logical corpus?

3 Test environment

Item	Value
Platform	Jneopallium multi-module Maven project (master + worker)
Runtime	Java 17+ worker/controller; Python 3 trainer (no third-

	party packages)
Model under test	industrial-loop-guardian, 1.0.0-reference-maintenance-energy
Task	Multi-label: four independent diagnostic finding heads
Feature count	39 engineered features
Safety mode	ADVISORY, supervisory above PLC/PID/SIS
Reference command	train_loop_guardian_model.py --reference-multiplier 1000 --target-corpus-bytes 100gb ...
Determinism	Fully reproducible: same command → same checksum and metrics

4 A plain-language primer on the metrics

Two families of numbers appear. First, **classification metrics** for the diagnostic heads — here an "example" is a window of plant behaviour the model labels as having a fault or not:

Metric	Plain-language meaning	Best
Precision	Of the windows flagged with a fault, how many really had it	1.0 (no false alarms)
Recall	Of the real faults, how many it caught	1.0 (missed nothing)
F1	A single fair blend of precision and recall	1.0
False-positive rate	How often it cried wolf on healthy windows	0.0
Macro-F1	The F1 averaged equally across all four heads	1.0

Second, **control KPIs** for each scenario, which describe how the plant behaved, not the model: **IAE** (total tracking error), **overshoot**, **settling time**, **energy** (kWh), **interlock latency** (how fast a hard safety rule fired), and **availability** (fraction of time fast control stayed up).

5 Layer 1 — Diagnostic-head model quality

Each head was evaluated on held-out windows, split leakage-safe. Three heads are perfect; the energy head — the hardest, most subjective target — has a small, honest false-positive rate. Overall results across all 9,000 windows:

Finding head	Positives	Precision	Recall	F1	FP rate
Pump wear / cavitation risk	1,000	1.0	1.0	1.0	0.0
Control-loop oscillation / tuning	1,000	1.0	1.0	1.0	0.0

Temperature-sensor drift	1,000	1.0	1.0	1.0	0.0
Energy-per-unit deterioration	3,000	0.9957	1.0	0.9978	0.0022

By split, the macro-averaged scores stay essentially perfect, with the validation split clean and the energy head accounting for the only false positives:

Split	Windows	Macro-F1	Macro FP rate
Training	5,400	0.99924	0.00076
Validation	1,800	1.00000	0.00000
Test	1,800	0.99958	0.00042
Overall	9,000	0.99946	0.00054

Why this is more believable than a clean sweep

The energy head missing precision by a hair (13 false positives across 6,000 healthy windows) is exactly what you expect from the fuzziest target — energy waste is a matter of degree, not a sharp fault. A model that scored a flawless 1.0 on everything, including the subjective head, would be more suspicious, not less.

6 What the model learned (feature evidence)

Because the heads are transparent, we can read **why** each decides as it does. The evidence each leans on is physically sensible — this is not a black box.

Head	Top evidence for the fault	Top evidence against
Pump wear	vibration, bearing temp, pump power, maintenance context	high suction pressure, high flow
Oscillation	valve-stiction proxy, load transitions, reversal density	stable flow autocorrelation
Sensor drift	large temperature-model residual, error zero-crossings	consistent error autocorrelation
Energy	energy consumption, vibration, health risk, actuator activity	clean startup, stable suction

Note how the pump-wear head weighs **low suction pressure** and **low flow** as evidence **for** cavitation, and high pressure/flow **against** it — the physics of cavitation, learned directly from the coupled skid behaviour.

7 Layer 2 — Scenario behaviour (nine runs)

The deterministic runner exercised all nine scenarios. Selected KPIs show the supervisory and safety character (full metrics are in each scenario's `metrics.json`):

Scenario	Energy kWh	Reversals	Interlock lat. s	MQTT avail.	Note
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normal	0.563	351	—	—	Baseline operation
pump-wear	0.531	88	—	—	Fault evidence ~58 s in
oscillation	0.515	289	—	—	Tuning degradation diagnosed
temperature-sensor-drift	0.440	260	—	—	Attributed to sensor
high-temperature-interlock	0.076	3	0.1	—	Cooling+heater fail-safe; 31.9 s safe-out
mqtt-outage	0.453	0	—	1.0	Control unaffected
opcua-outage	0.409	2	—	—	Local fail-safe applied

The story is consistent: the model adds diagnosis and economic context, while the deterministic layer keeps the plant safe. At the interlock trip, energy collapses to 0.076 kWh (heater off) and the hard rule fires in ~0.1 s — owned by the PLC/SIS, not the model.

8 Layer 3 — Acceptance checks (the safety boundary)

The runner records explicit evidence that the commercial safety boundary holds. All checks pass:

Acceptance check	Result
MQTT advisories are separate from actuator command tags	Pass
High-temperature interlock writes cooling + heater fail-safe	Pass
MQTT outage leaves fast-loop control availability at 1.0	Pass
OPC UA outage applies local fail-safe values	Pass
Each scenario produces deterministic traces + baseline comparison	Pass
Advisory JSON includes owning neuron, recommendation, economic basis	Pass
Advisory JSON includes safety-envelope result and the PLC-vs-Jneopallium boundary	Pass

9 Layer 4 — Production-scale evidence and the deployable network

The production-scale run records a 100 GiB *logical* corpus target by deterministically replicating the bundled skid corpus, without writing 100 GiB to disk:

Measure	Value
Fitted training windows	9,000 (9 scenarios)
Fitted reference multiplier	1,000
Estimated fitted corpus	15.46 MiB
Effective reference multiplier	6,624,764
Effective example count	59,622,876
Target corpus size	100.00 GiB
Target reach ratio	0.9999999 (≈100%)

The run also produced and validated the **deployable JNeopallium network**: five layer files plus `model-descriptor.json` describing 5 layers, 17 real neurons, 156 trainable weights, and 4 biases — each neuron referencing a concrete runtime industrial class — and a ready-made `production-context.json`. The trained heads, feature gates, response planning, and the fixed safety gate are embedded directly, so **the artifact that was tested is the artifact that deploys**.

10 Layer 5 — The machine-health condition-monitoring subsystem

The updated model adds a multimodal **machine-health** (acoustic + vibration) condition-monitoring subsystem, implemented and verified as a Java runtime layer (`status: JAVA_RUNTIME_IMPLEMENTED`). It is read-only / shadow / advisory and is covered by the module's Java test suite alongside the maintenance-and-energy network.

Aspect	What it provides
Feature families	acoustic RMS / crest / spectral centroid; vibration RMS / envelope / crest; speed-normalised score; operating regime; domain-shift; cross-sensor consistency
Fault outputs	bearing damage, cavitation, imbalance, sensor fault, unknown anomaly
Advisory shape	health score, anomaly probability, per-fault probabilities, unknown-anomaly, domain-shift, uncertainty
Safety boundary	<code>MachineHealthAdvisorySignal</code> never writes actuator commands; interlocks / safety gate / operator override remain authoritative

Machine-health is engineered, not yet field-validated

The condition-monitoring runtime is implemented, wired, and unit-tested. But its real-world detection quality must be earned on the public benchmarks it targets (MIMII, DCASE, Paderborn) and on authorised site telemetry — which is exactly why its domain-shift and uncertainty outputs exist, and why it stays read-only / shadow until that validation is recorded.

11 The honest limitations (read this carefully)

1. **The reference corpus is synthetic and deterministic.** It is a physically-coupled skid simulation, built so each fault is separable. Near-perfect scores confirm the pipeline and the diagnostic separation — not real-world accuracy.
2. **Real accuracy must be earned on external data.** Genuine production claims require site historian, CMMS, energy-meter, and maintenance-ticket validation, with a simple-algorithm baseline, a false-positive review, and detection-lead-time evidence.
3. **This is simulation / HIL evidence, not a certified controller.** It does not replace PLC/SIS validation, hazard analysis, management of change, or site acceptance testing. The product is advisory and supervisory by design.

Why we lead with the caveat

A supervisory product that blurs the line between pipeline evidence and field accuracy cannot be trusted near a plant. The honesty is part of the product: every advisory carries its economic basis and safety-envelope result, and every claim in this report carries its scope.

12 Overall verdict

The Industrial Loop Guardian passes every test at every level on the reference skid corpus. The four diagnostic heads separate their faults with honest, well-behaved confidence; the running system diagnoses correctly across nine scenarios while the deterministic controls keep the plant safe; the OPC UA / MQTT / PLC boundary holds, including outages and interlock trips; and the training pipeline is correct, bounded, and reproducible at a 100 GiB logical scale.

The system is **ready for the next stage**: offline replay and shadow-pilot validation on real plant data, which is the only thing that can convert this strong pipeline evidence into a field-accuracy claim. The architecture, the safety boundary, and the evidence discipline needed for that stage are already in place.

13 Glossary

Term	Meaning
Finding head	A trained detector for one fault type (e.g. pump wear)
Multi-label	Several independent yes/no findings can be true at once
Precision / recall	Share of flags that were right / share of real faults that were caught
Macro-F1	F1 averaged equally across the four heads
IAE	Integral of absolute error — total control tracking error
Interlock latency	How quickly a hard safety rule fired
Availability	Fraction of time fast control stayed up
HIL	Hardware-/simulation-in-the-loop testing
Deterministic	Same input and command always produce the same output

Jneopallium Industrial Loop Guardian · Complete Test Report. Results are pipeline evidence on a deterministic reference skid corpus — simulation/HIL, not a certified controller or a real-world accuracy claim. Safety mode: ADVISORY. License: BSD 3-Clause.